Cairo University

Faculty of Computers and Artificial Intelligence

CS251

Introduction to Software Engineering

$penderman

Software Design Specifications

Version 1.1

Apr-2026

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Project: $penderman

Software Design Specification

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Software Design Specification

Team

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Document Purpose and Audience:

Document Purpose and Audience

Document Purpose

This Software Design Specification (SDS) serves as a comprehensive **technical blueprint** for the **Spenderman** personal budgeting application. It translates functional requirements into a structured architectural design to guide the development and testing phases.

Key objectives of this document include:

- **Defining System Scope:** Outlining the boundaries of the application and its primary interactions with the "System User".
- **Structural Architecture:** Detailing the high-level technical containers, including the **JavaFX GUI**, the **Java Back-end**, and the **Microsoft SQL Server** database.
- **Behavioral Logic:** Providing step-by-step logic for critical operations such as user registration, transaction processing, and financial goal creation.
- **Data Integrity:** Establishing how financial data is persisted and retrieved through secure communication between the back-end and database layers.

Target Audience

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Software Design Specification

This document is intended for stakeholders involved in the technical execution and management of the project:

Audience	Responsibility
Software Developers	To implement the specific classes, methods, and logic flows defined in the sequence diagrams.
Project Managers	To verify that the technical design aligns with the project's scope and business goals.
Software Architects	To ensure the structural integrity of the container-level interactions and technology stack.
Quality Assurance (QA)	To design test cases based on the documented logic for transactions and user management.
Database Administrators	To manage the SQL Server schema required to support the application's data persistence.

System Models

I. Architecture Diagram

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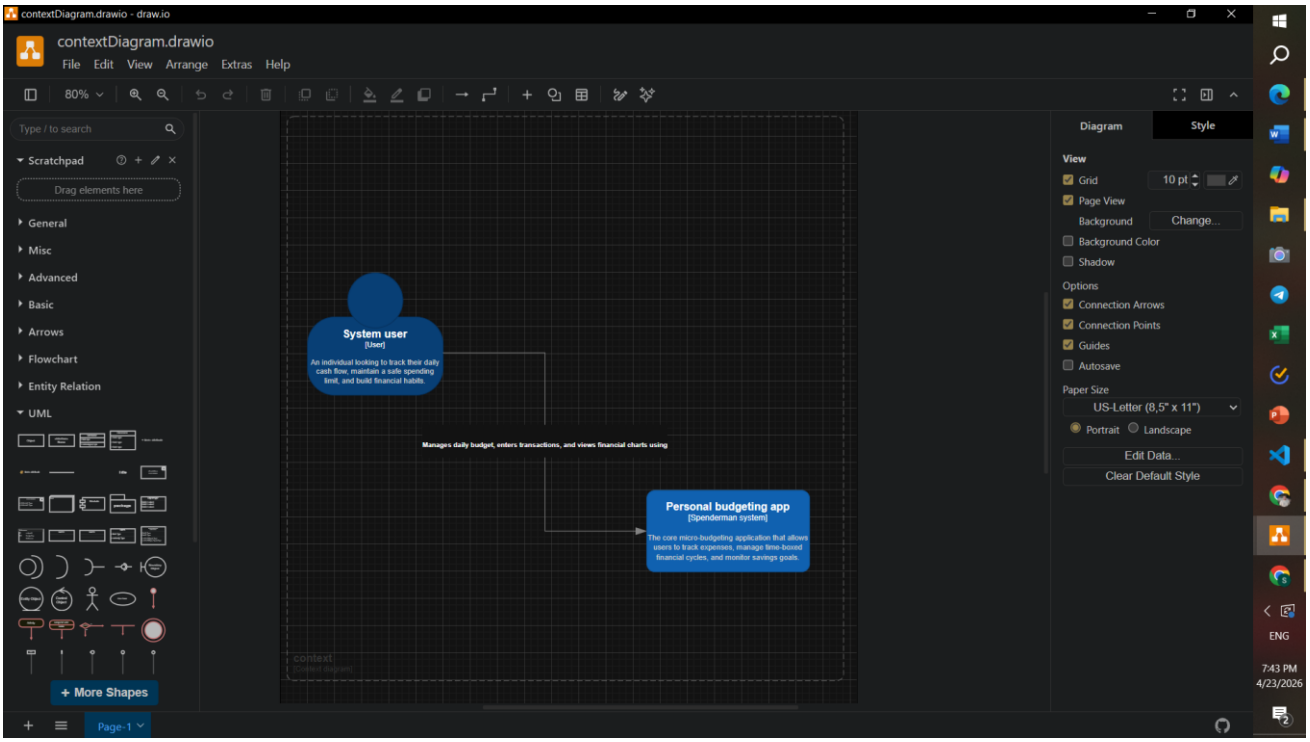

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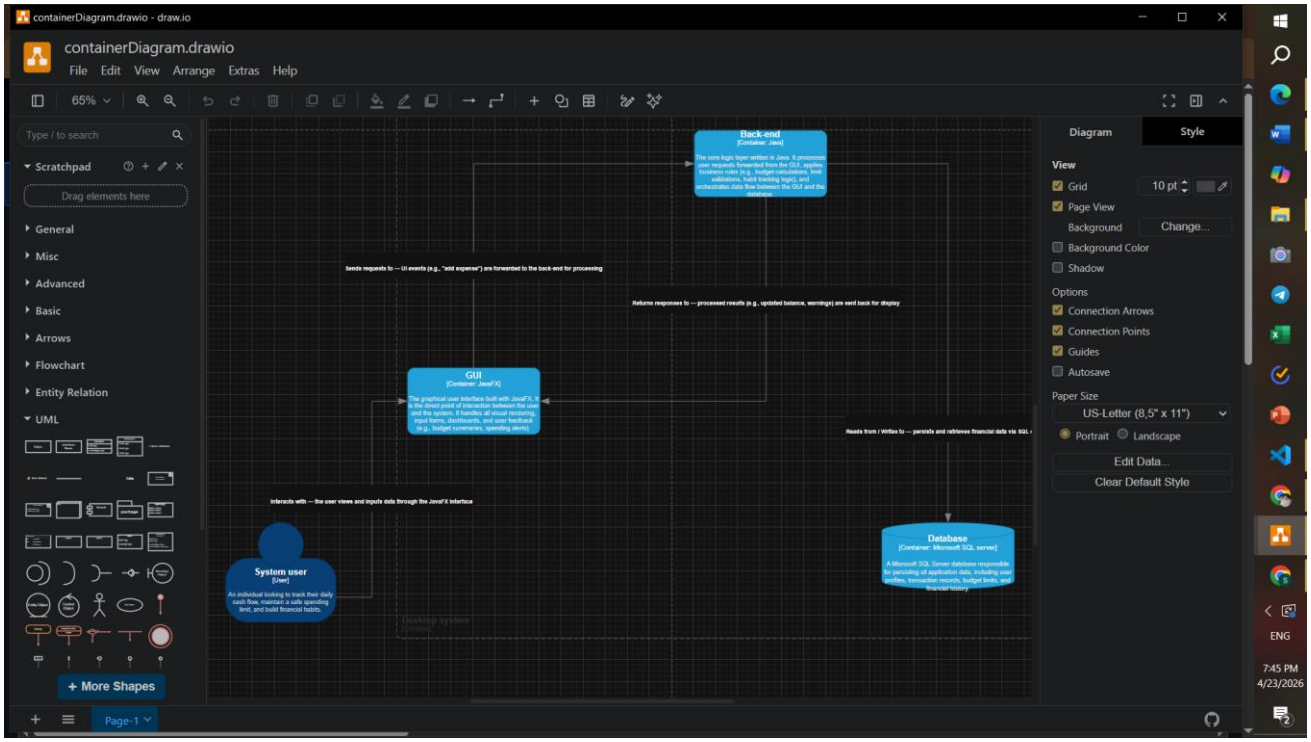

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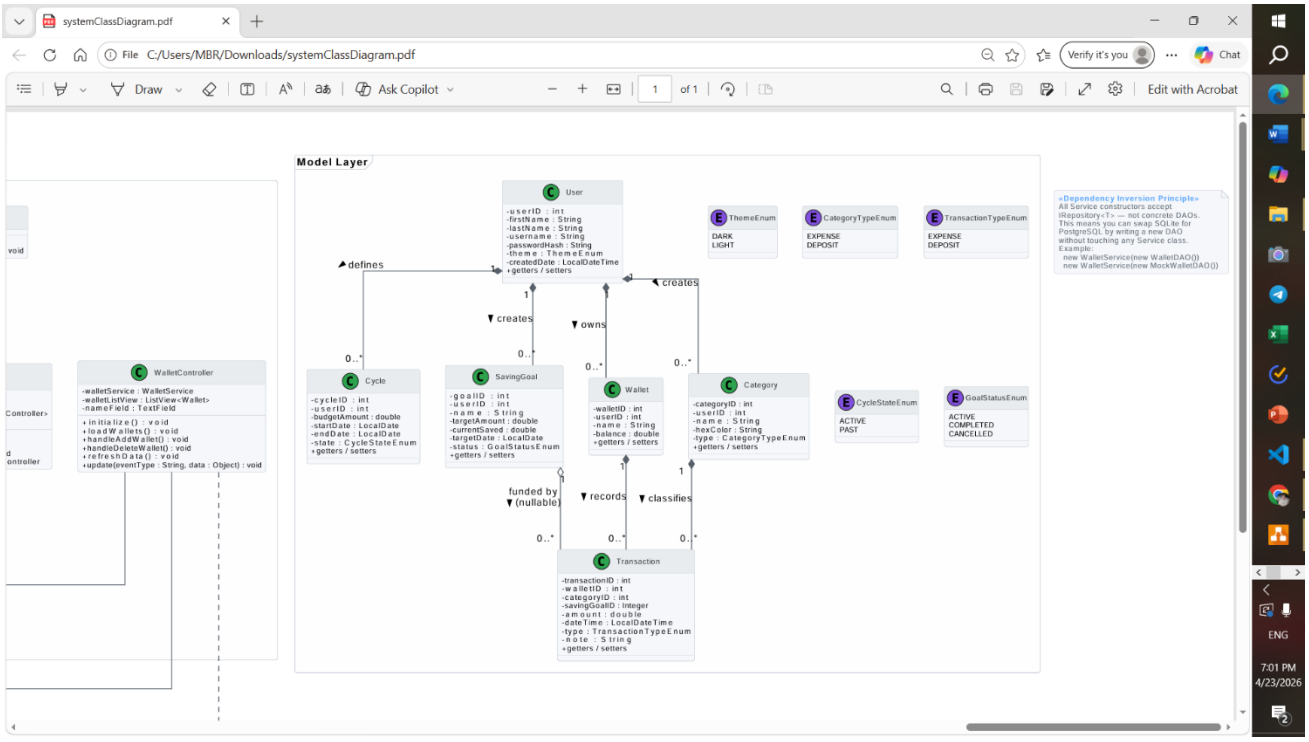
II. Class Diagram(s)



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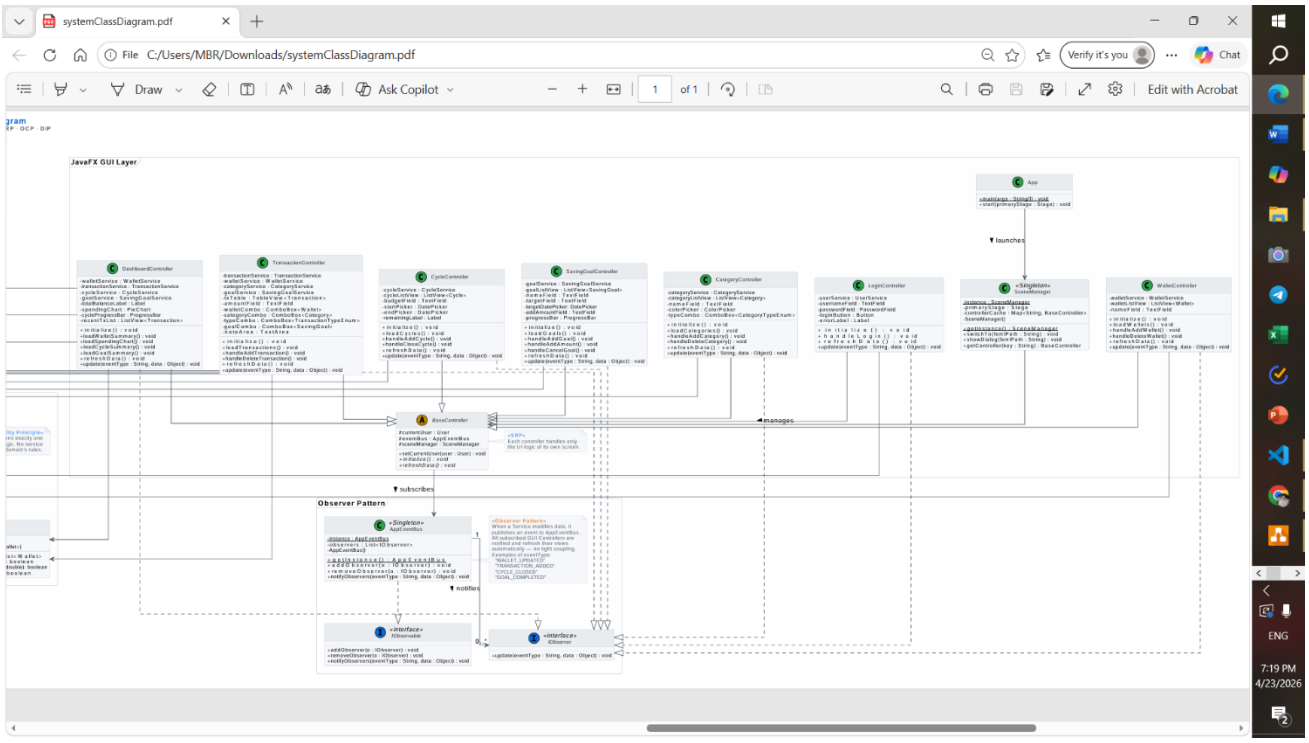

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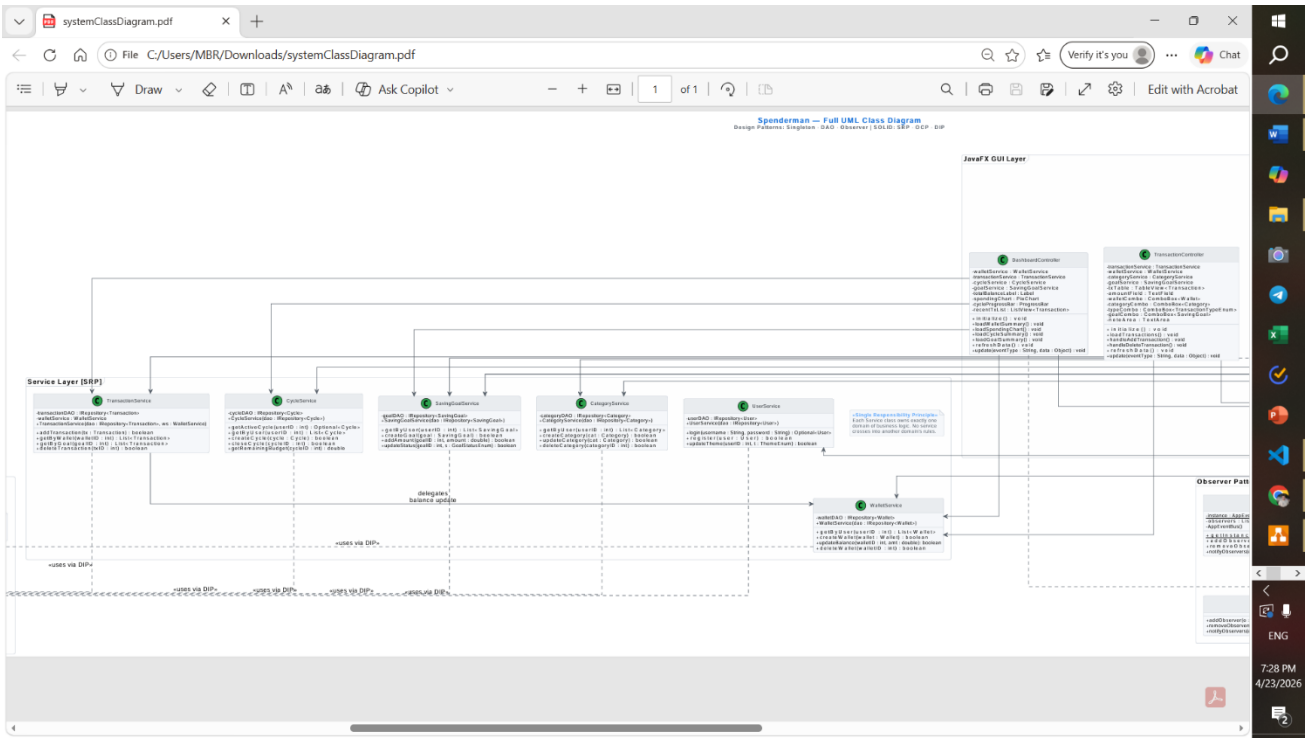

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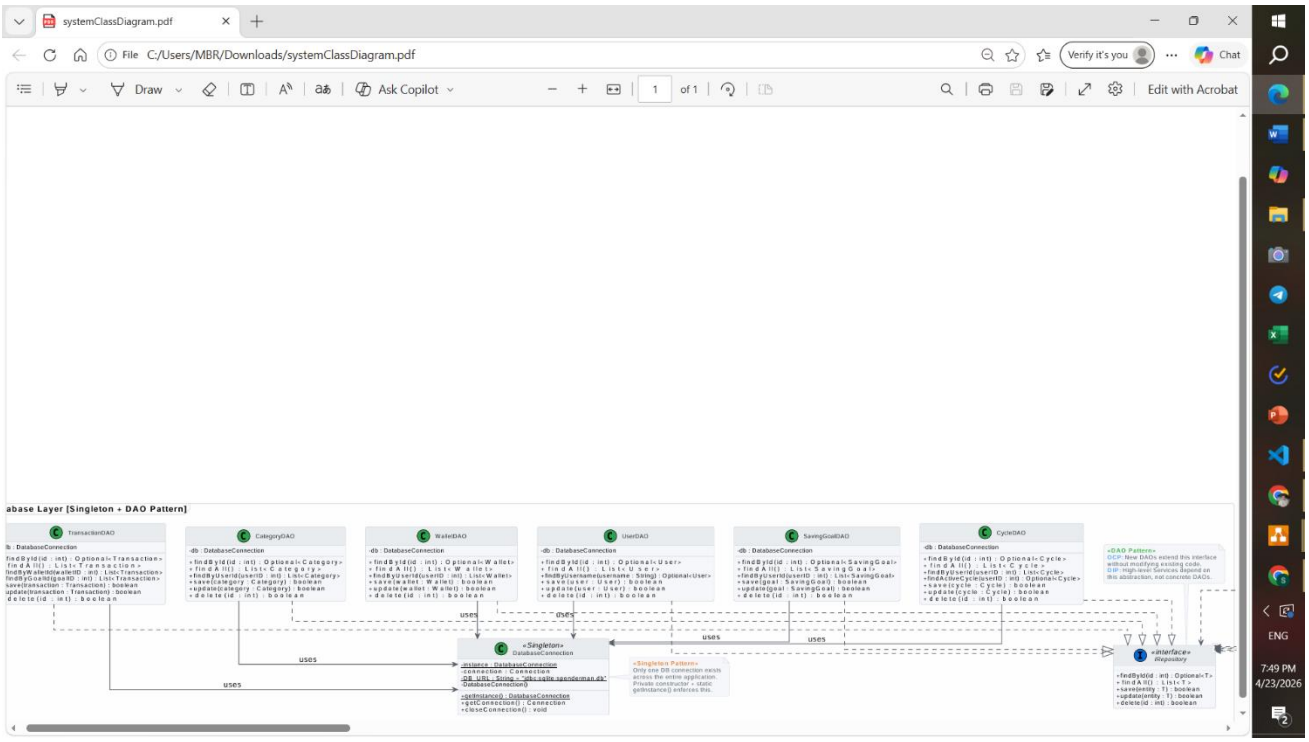

Software Design Specification

III. Class Descriptions

ID	Class Name	Description & Responsibility
1	User	Model: Represents the system user. Responsible for holding user credentials and profile data.
2	Wallet	Model: Represents a financial account. Responsible for holding the current balance and name.
3	Transaction	Model: Represents a single movement (amount, date, type). Links to Wallet and Category.
4	Category	Model: Represents classifications like Food, Rent, or Salary for organization.
5	Cycle	Model: Financial tracking period. Holds budget amount, dates, and active/past state.
6	SavingGoal	Model: Tracks target amounts, current savings, and deadlines.
7	DatabaseManager	Singleton: Manages the single database connection to prevent memory leaks.
8	GenericDAO	Data Access: Interface defining standard CRUD (Create, Read, Update, Delete) operations.
9	Entity DAOs	Data Access: Executes specific SQL queries (Insert, Update, Select) for each entity.
10	DashboardController	UI Layer: Handles main screen interactions and visual updates like Pie Charts.
11	WalletsController	UI Layer: Manages UI forms for creating, displaying, and deleting wallets.
12	AddTransactionController	UI Layer: Captures user input for transactions and passes it to the service layer.
13	CyclesController	UI Layer: Manages UI for setting budget limits and cycle dates.
14	TransactionService	Business Logic: Executes rules for expenses/deposits and deducts from wallets.
15	CycleService	Business Logic: Calculates safe daily limits and checks budget alerts.
16	DashboardService	Business Logic: Aggregates system-wide data (total balance, grouping expenses).
17	WalletService	Business Logic: Governs the rules for wallet creation and deletion.

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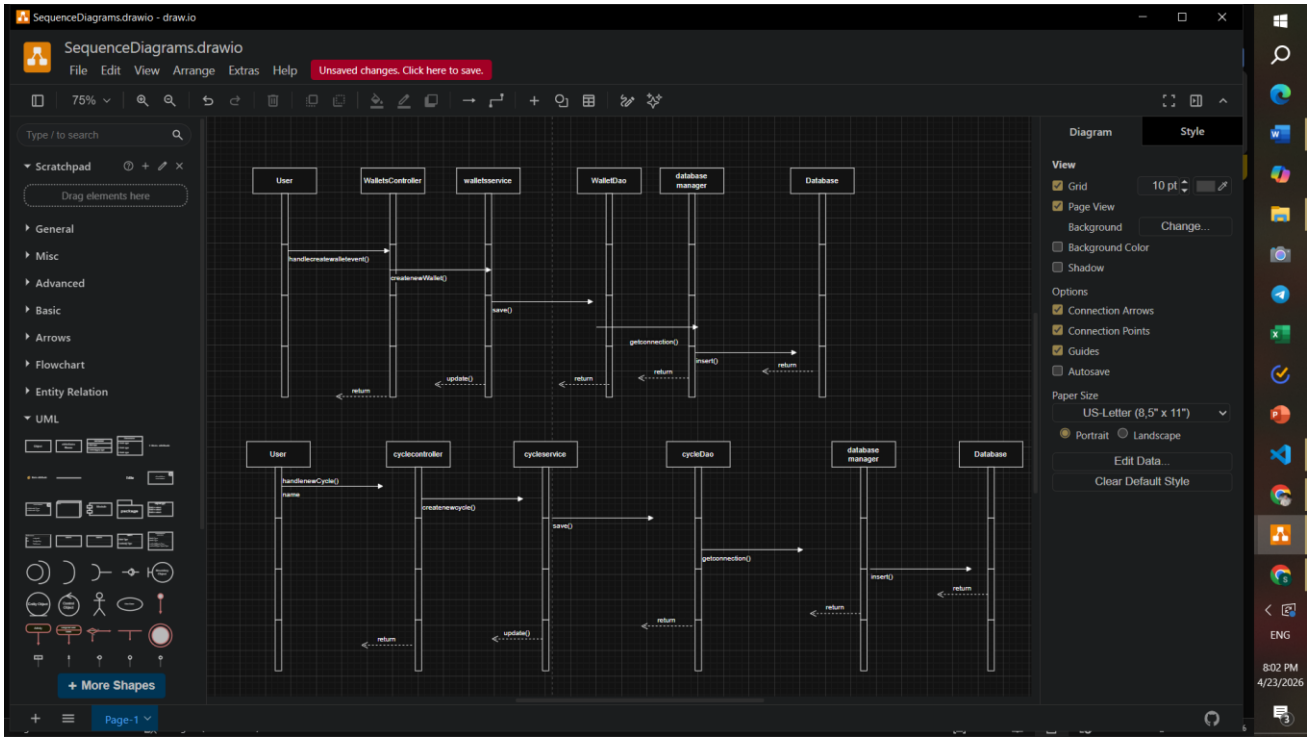
IV. Sequence diagrams

1-CREATE WALLET&&CREATE CYCLE



Visualize Cateogries:

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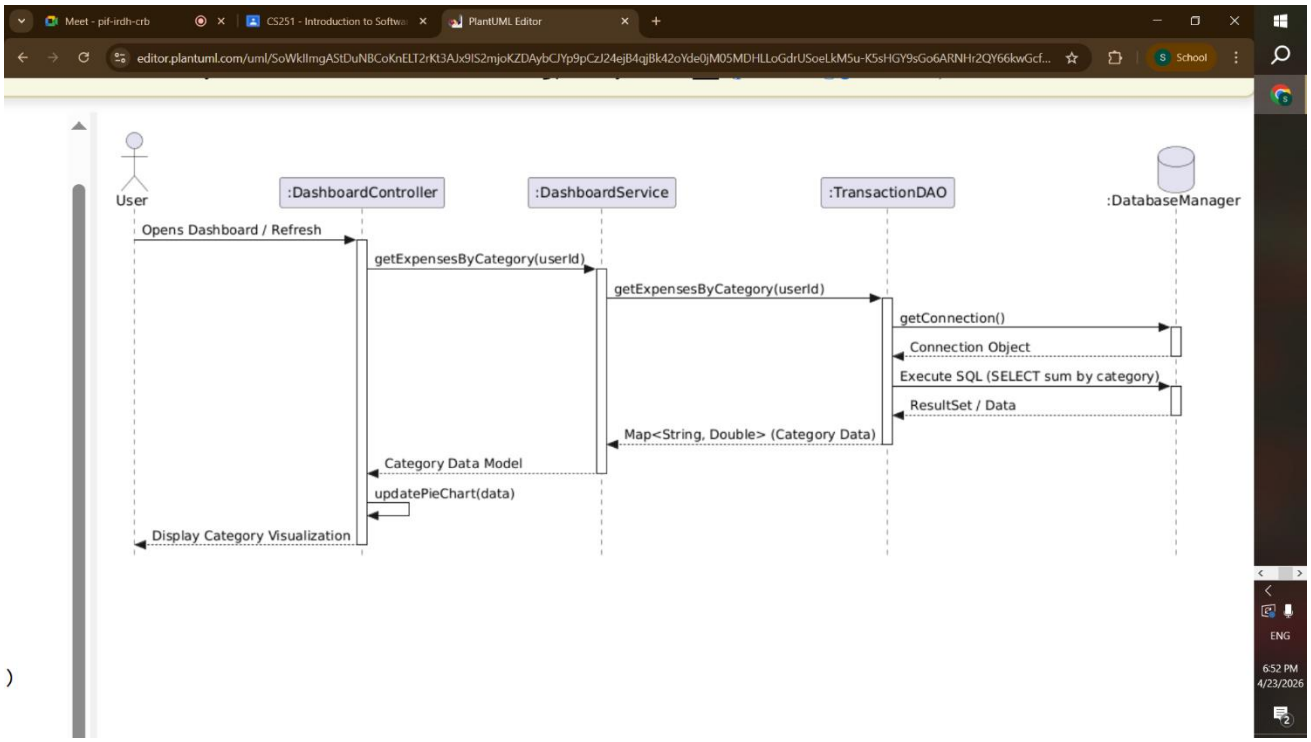

ADD TRANSACTION(EXPENSE):

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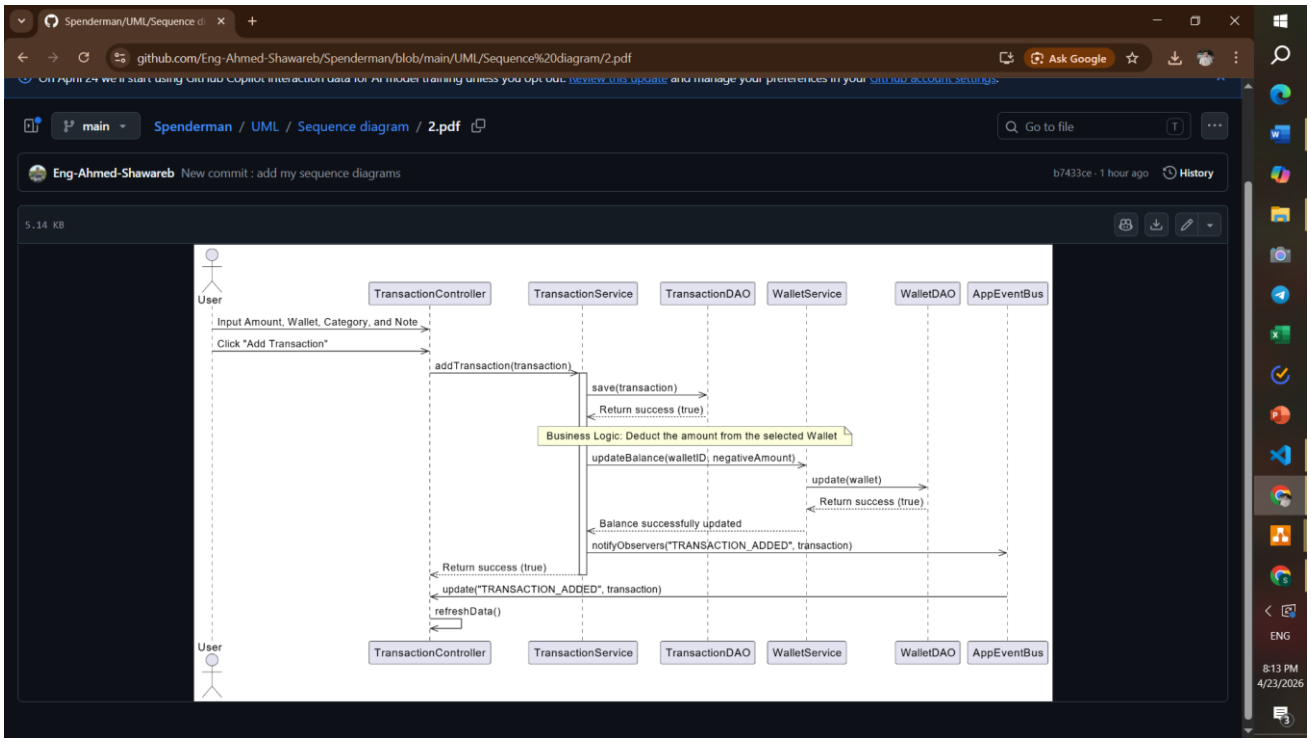

Signup

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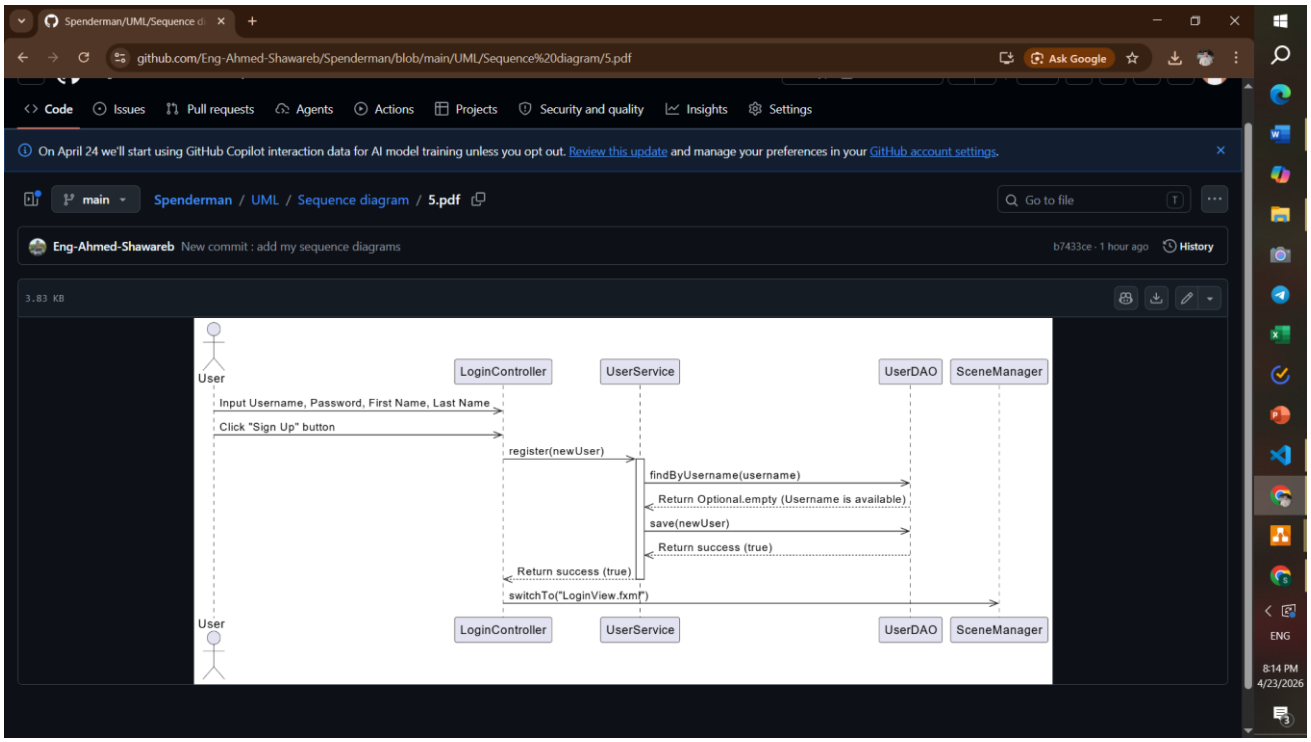
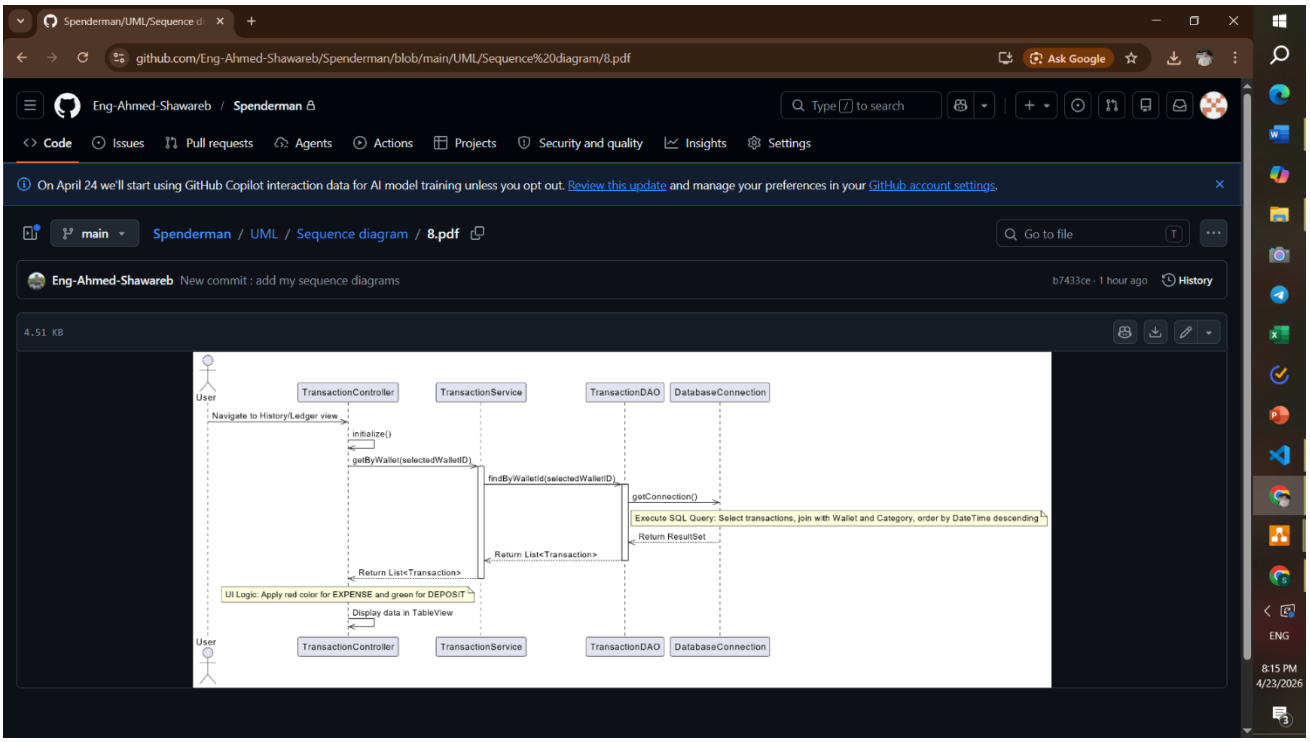


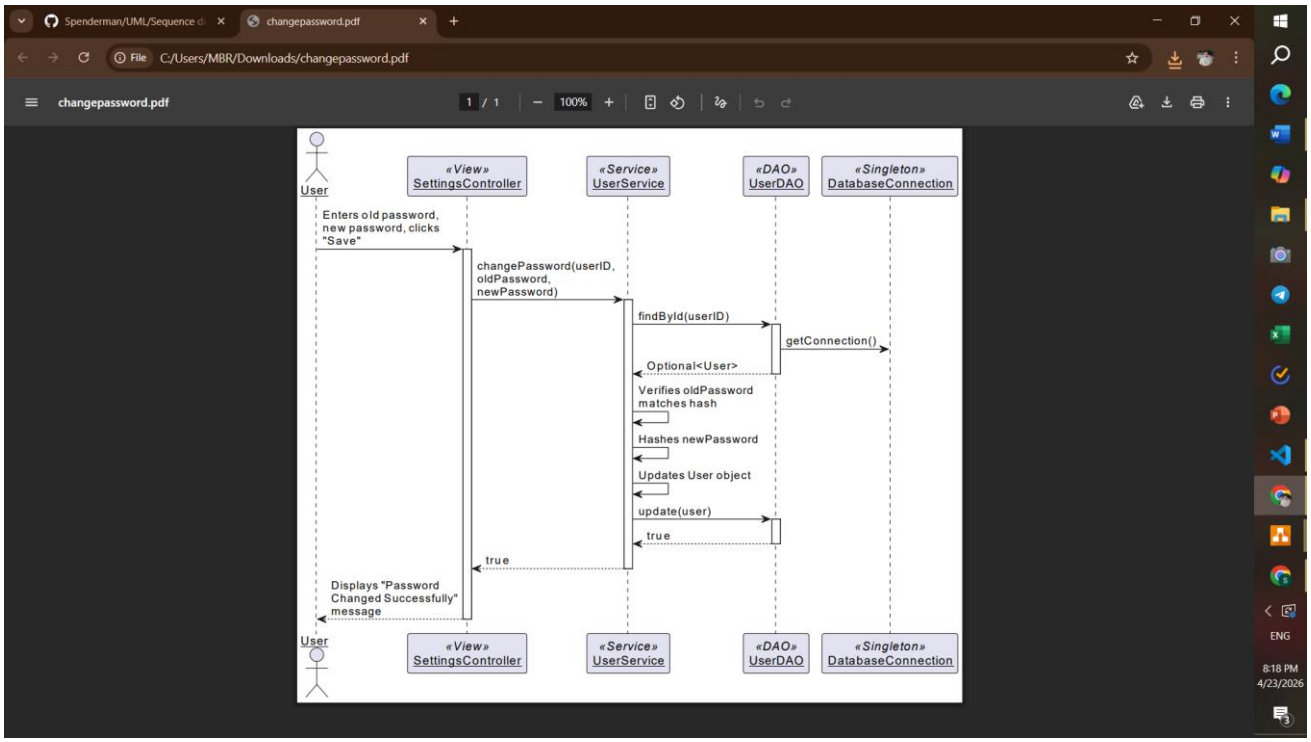
VIEW TRANSACTION HISTORY:

Software Design Specification



CHANGE PASSWORD:

Software Design Specification



CREATE DEPOIST:

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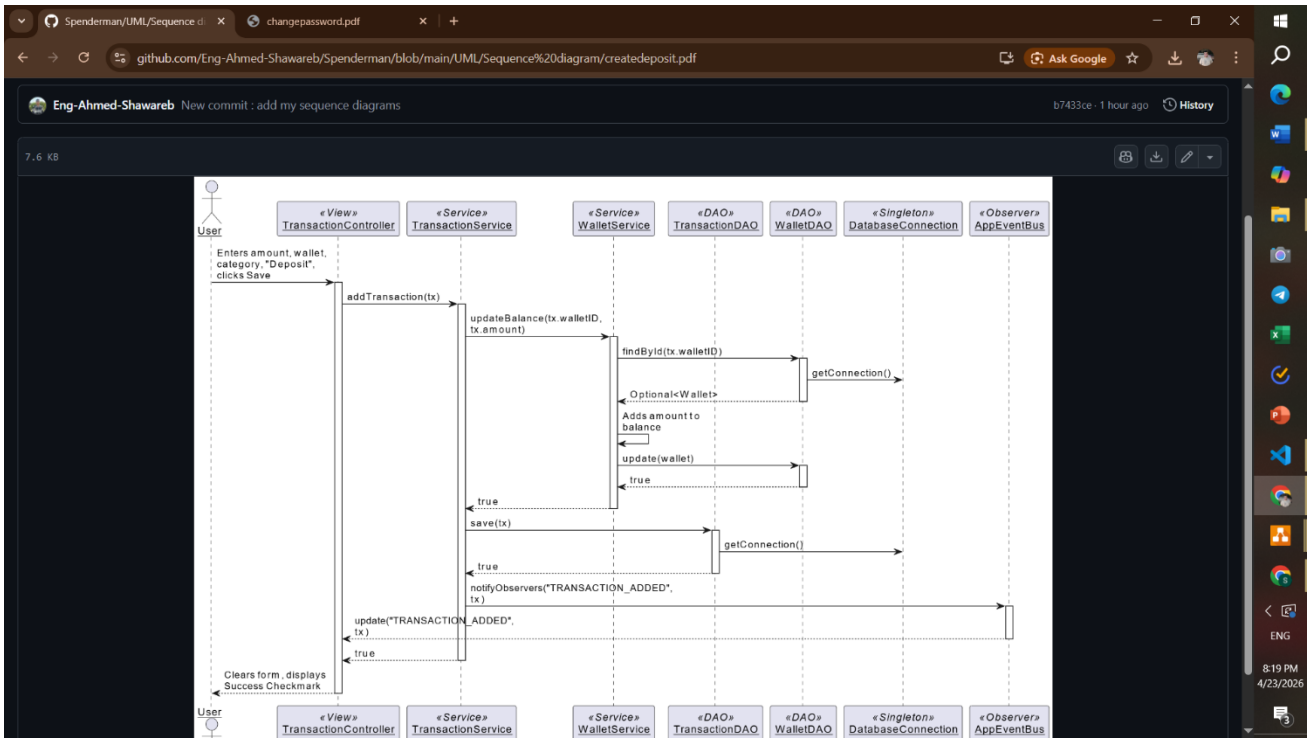

CREATE SAVING GOALS:

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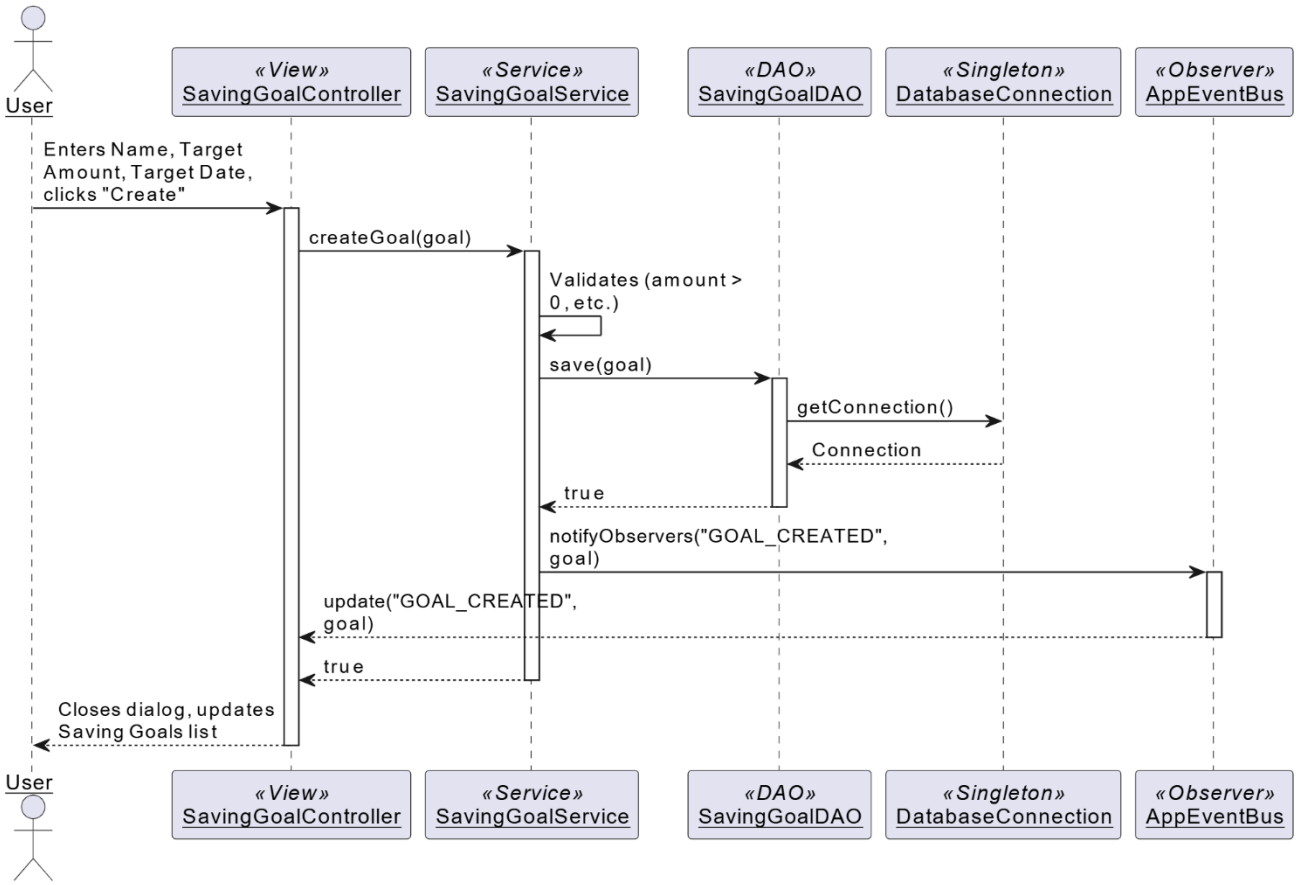

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Sequence Diagram	Classes Used	All Methods Used
1. Add Transaction (Expense)	User, TransactionController, TransactionService, TransactionDAO, WalletService, WalletDAO, AppEventBus	addTransaction(), save(), updateBalance(), update(), notifyObservers(), refreshData()
2. signup	User, LoginController, UserService, UserDAO, SceneManager	register(), findByUsername(), save(), switchTo()
3. Create Wallet	User, WalletController, WalletService, WalletDao, DatabaseManager, Database	handleCreateWallet(), createNewWallet(), save(), getConnection(), insert()
4. Create Cycle	User, CycleController, CycleService, CycleDao, DatabaseManager, Database	handleNewCycle(), createNewCycle(), save(), getConnection(), insert()
5. Visualize cateogries	User, DashboardController, DashboardService, TransactionDAO, DatabaseManager	getExpensesByCategory(), getConnection(), updatePieChart()
6. Change Password	User, SettingsController, UserService, UserDAO, DatabaseConnection	changePassword(), findById(), getConnection(), update()
7. Add Transaction (Deposit)	User, TransactionController, TransactionService, WalletService, TransactionDAO, WalletDAO, DatabaseConnection, AppEventBus	addTransaction(), updateBalance(), findById(), getConnection(), update(), save(), notifyObservers()
8. Create Saving Goal	User, SavingGoalController, SavingGoalService, SavingGoalDAO, DatabaseConnection, AppEventBus	createGoal(), save(), getConnection(), notifyObservers(), update()
9. View Transaction History	User, TransactionController, TransactionService, TransactionDAO, DatabaseConnection	initialize(), getByWallet(), findByWalletId(), getConnection()

V. State Diagram

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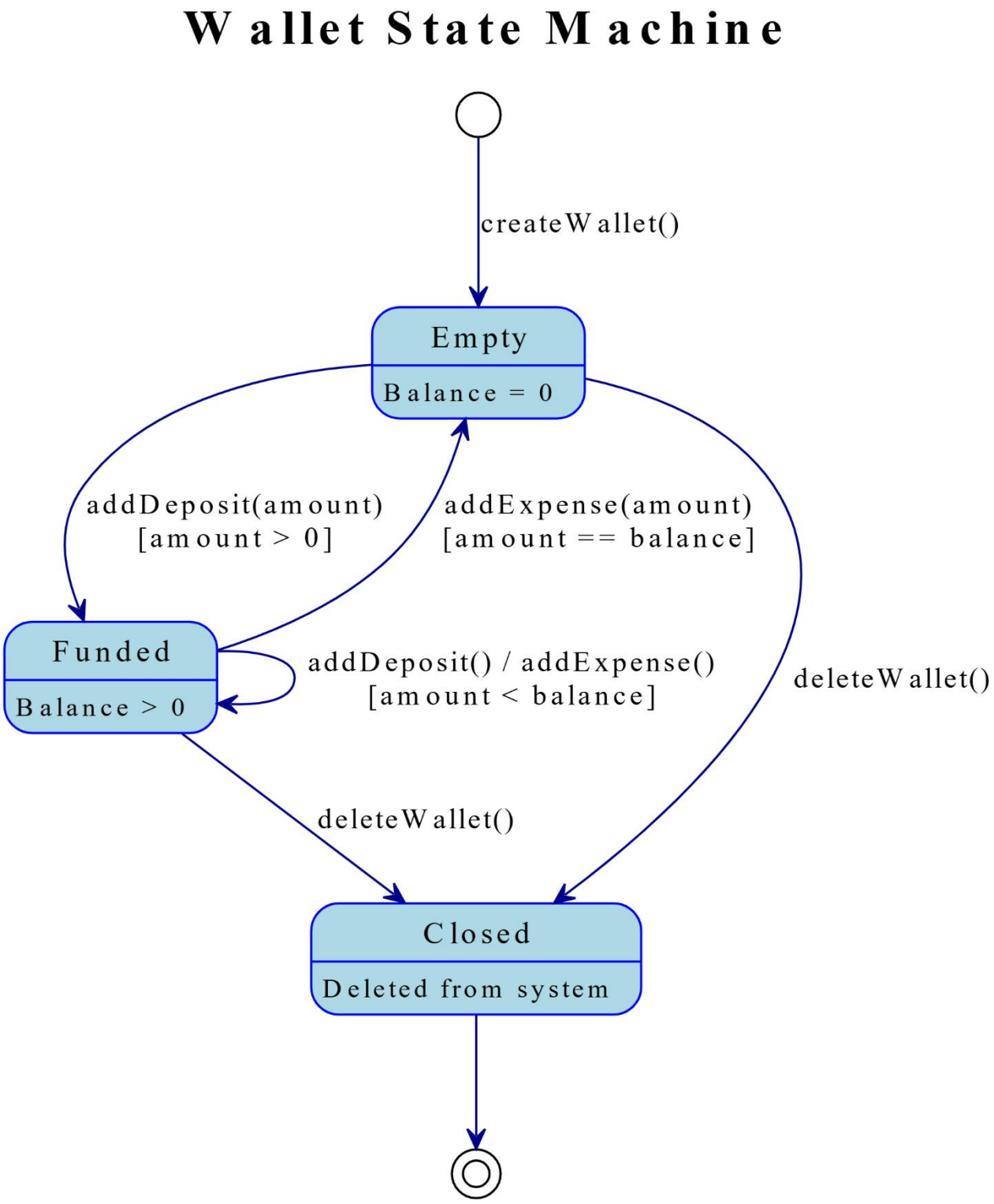

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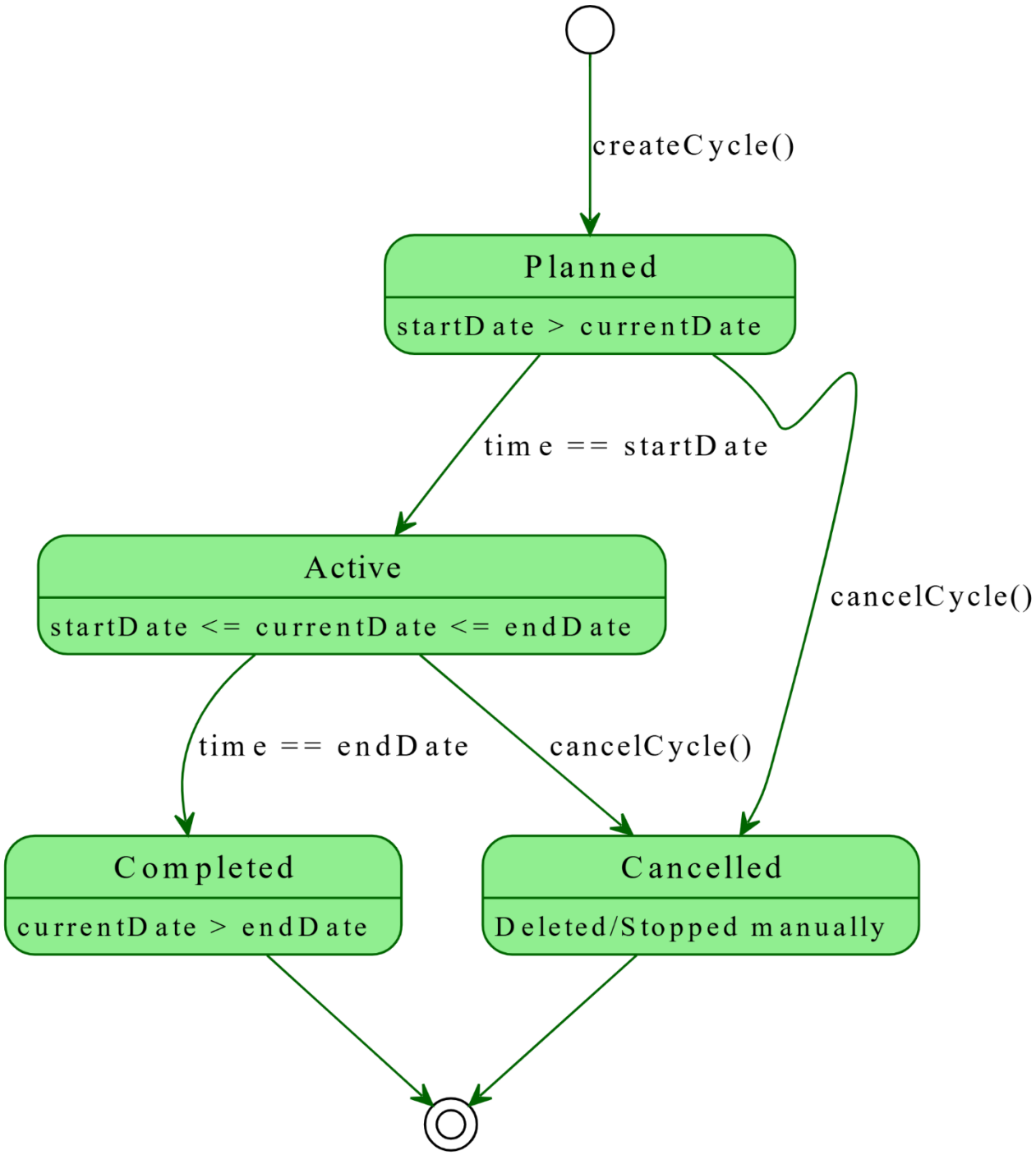
Cycle State Machine



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DATABASE DIAGRAMS:

ER DIAGRAM:

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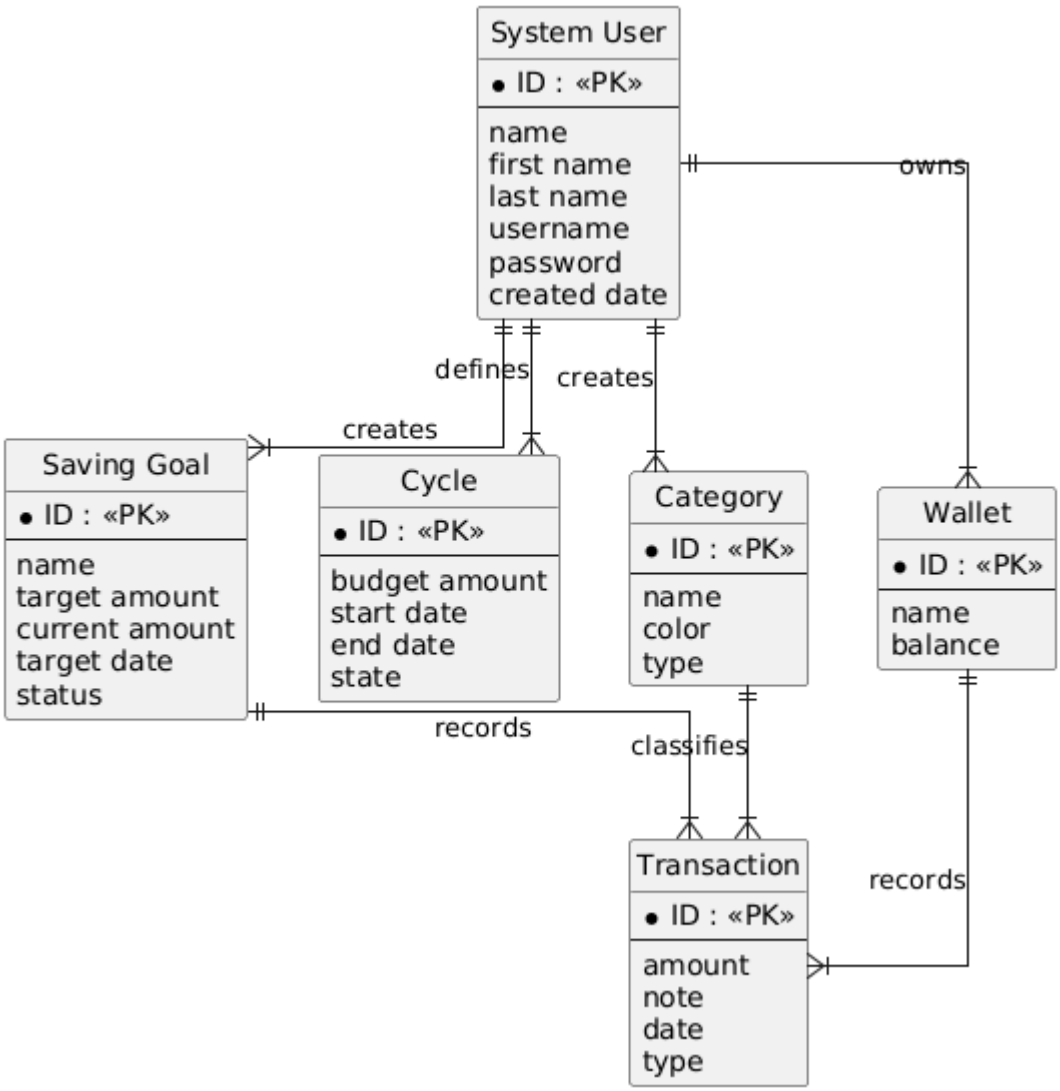

Schema:

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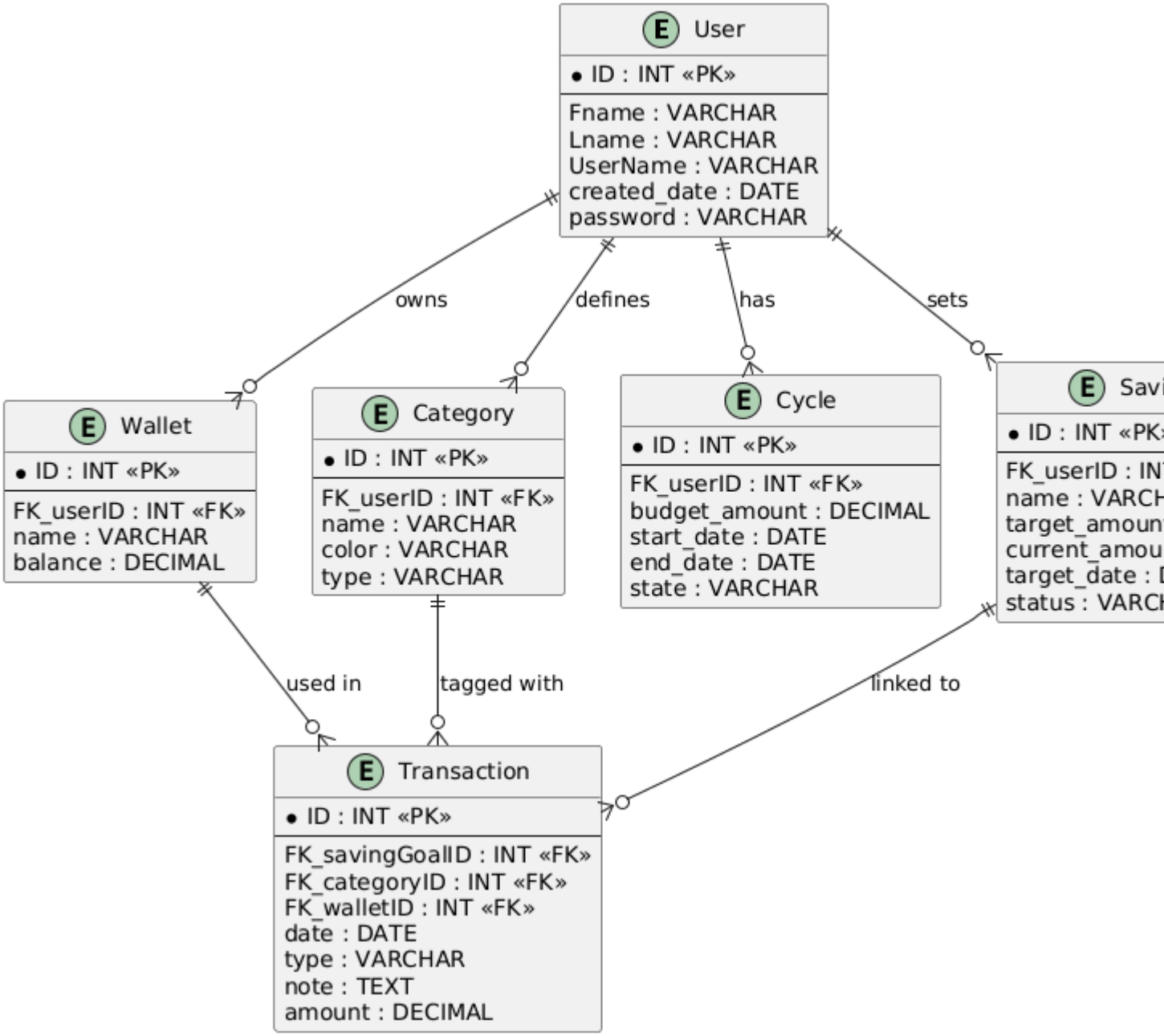

VI. SOLID Principles

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Software Design Specification

Written in class diagram.

VII. Design Patterns

Written in class diagram .

Tools

PlanetUML, Figma, drawio.

Ownership Report

owner	Items
Ahmed Amr 20240038	**Er diagram,3seq diagram,schema,uml**
Mohamed Ayser 20240476	*Uml ,state diagram,3seq,er diagram*
Ahmed Mohamed 20240049	*Ui ,uml ,er diagram,3seq*